

Shiloh National Military Park

Shiloh, Tennessee

REPLACING FIXTURES AND TILE IN
COMFORT STATION
PMIS NO. 331321

PROJECT SPECIFICATIONS



NATIONAL PARK SERVICE
SHILOH NATIONAL MILITARY
PARK
FEBRUARY 7, 2024

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SECTION 010100

DESCRIPTION OF WORK

PART 1 GENERAL INFORMATION

1.1. GENERAL: The work of this section summarizes the work involved in this Contract and explains the contractor's responsibility in maintaining the condition of the premises.

1.2. SUMMARY:

- A. The work of this contract consists of the general renovation of the Comfort Station at the Shiloh National Military Park as provided in these Specifications.
- B. All work will be performed under a single contract.
- C. All work shall conform to the International, Tennessee and local Building Codes and their amendments, the ABA/ABAAS Accessibility Guidelines, and NPS regulations.
- D. The Comfort Station is an historic building making this project subject to all historic standards by the Department of Interior. The contractor must therefore adhere to the Secretary of Interior's *Standards for the treatment of Historic Properties: Secretary of Interior's standards for Rehabilitation*. Rehabilitation is defined in 36 CFR 67 as "the process of returning a property to a state of utility, through repair or alteration, which makes possible an efficient contemporary use while preserving those portions and features of the property which are significant to its historic, architectural, and cultural values."

SCOPE: The scope of work includes but is not limited to the that described below, which is integral to the information provided in these Specifications and the other Contract Documents.

1.3.

A. Comfort Station (Interior): Demolition of partitions, plumbing fixtures, approximately 1200 square feet of ceramic wall and floor tile (size and color to be determined by the Park), electric hand dryers, sinks and vanities. Installation of new vanities to match existing, new plumbing fixtures to include sinks, on demand hot water for sinks, urinals, and commodes including new Zurn or equivalent valves that meet EPA WaterSense standards. New ceramic floor and wall tiles with size and color to be determined by NPS COR. Floor tile will meet Architectural Barriers Act (ABA) standards for non-slip. New solid surface partitions, vanity tops, and hardware. New faucets and electric hand dryers. New paint on walls and ceiling not covered by tile, color to be determined by NPS COR.

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1.4. LOCATION: Shiloh National Military Park, Shiloh Tennessee



1.5. DEFINITION OF BID ITEMS:

- A. BASE BID: Replacement of interior fixtures in the Comfort Station.

- 1.6. SITE VISIT: The Contractor is urged and expected to inspect the site where services are to be performed and to satisfy themselves regarding all general and local conditions that may affect the cost of contract performance, to the extent that the information is reasonably obtainable. In no event shall failure to inspect the site constitute grounds for a claim after contract award.
- 1.7. CONSTRUCTION SEQUENCE: The construction sequence shall be as required to perform all the Work of these Specifications and Drawings in the most efficient and timely manner.
- 1.8. PERIOD OF PERFORMANCE: The Period of Performance shall be for within four months of the Notice to Proceed.
- 1.9. PHYSICAL SECURITY: The contractor shall be responsible for safeguarding all equipment, information, and property at the location. At the close of each work period, government facilities, equipment, and material shall be secured.
- 1.10. GOVERNMENT FURNISHED PROPERTY, EQUIPMENT, AND OF PREMISES
 - A. Facilities: The Government shall provide parking and a construction staging area during the period of work.
 - B. Utilities: The Government will provide electrical power to the site. The Contractor shall instruct employees in utilities conservation practices. The Contractor shall be responsible for operating under conditions that preclude the waste of utilities, which include turning off the water faucets or valves after using the required amount to accomplish cleaning vehicles and equipment.
- 1.11. CONTRACTOR'S USE OF PREMISES
 - C. Construction Camp: The Contractor shall maintain work within the established construction site and always conduct operations to ensure the least inconvenience to NPS staff, property, and visitors. Establishment of camping facilities shall not be permitted or any parking outside the immediate job site, or as directed by the Contracting Officer.
 - D. Storage of materials shall be within the job site as approved by the Contracting Officer.
 - E. Preservation of Natural Features: Confine all operations to work limits of the project. Prevent damage to natural surroundings. Restore damage areas, repairing or replacing damaged trees and plants, at no additional expense to the Government.
 - F. Existing site features, including walls, fences, paving, curbing and other construction, not scheduled for work, shall be protected so as not to be damaged due to the contractor's activity. Any damage that does occur to these areas shall be returned to the pre-contract condition as part of this contract at no additional cost to the Government. Local patching to correct damage will not be acceptable in cases where patching will be recognized. Entire regional replacement/restoration will be required.

- G. Existing Utilities: Contractor shall be responsible for locating and preventing damage to known utilities. If damage occurs, repair utility at no additional expense to the Government.
- H. Hauling Restrictions: Comply with all legal load restrictions in the hauling of materials and equipment.
- I. Work Hours: The contractor shall schedule the work only during the hours of 7 am to 5 pm Monday through Friday. Requests for any work outside those hours, including weekends and holidays, must be submitted for approval at least 48 hours in advance.
- J. Holidays:

New Year's Day	Labor Day
Martin Luther King Jr's Birthday	Columbus Day
President's Day	Veteran's Day
Memorial Day	Thanksgiving Day
Independence Day	Christmas Day

- 1.12. FIELD VERIFICATION: Field verify all new and existing dimensions affecting the work of this contract before ordering products. Any discrepancies construction specifications, and field conditions shall be brought to the attention of the Contracting Officer before work is to be performed. Any work that is performed without notifying the Contracting Officer will be done at the contractor's own expense.
- 1.13. CONSTRUCTION MATERIALS: All materials shall be Contractor-furnished.

SECTION 011700

ACCIDENT PREVENTION

PART 2 GENERAL

1.14. DESCRIPTION: The work of this section consists of establishing an effective accident prevention program and providing a safe environment for all personnel and visitors.

1.15. SUBMITTALS:

E. Accident Prevention Program: Before on site work begins, submit for approval an accident prevention program. The Contracting Officer will review the proposed program for compliance with OSHA and project requirements. If the program requires any revisions or corrections, the contractor shall resubmit the program within 10 days. No progress payments will be made until the program is approved. The program shall include:

- a. Name of responsible supervisor to carry out the program.
- b. Weekly safety meetings.
- c. First aid procedures.
- d. Outline of each phase of the work, the hazards associated with each major phase, and the methods proposed to ensure property protection and safety of the public, National Park Service personnel, and Contractor's employees. Identify the work included under each phase by reference to specification section or division numbers.
- e. Training, both initial and continuing.
- f. Planning for possible emergencies, such as floods, fires, cave-ins, explosions, power outages, and wind storms. Such planning shall take into consideration the nature of construction, site conditions, and degree of exposure of persons and property.
- g. Housekeeping and Fire Protection:

F. Before on-site work begins, submit for approval a Safety Plan. The Contracting Officer will review the proposed plan for compliance with park and project requirements. If the plan requires revisions or corrections, the contractor shall resubmit the plan within 10 days. The plan shall include:

- a. The name of the responsible supervisor to carry out the program.
- b. Outline methods and procedures for securing or removing materials at the job site in the event of an emergency.

G. Upon request, submit proof of employees' qualifications to perform assigned duties in a safe manner.

1.16. QUALITY ASSURANCE:

A. In case of conflicts between Federal, state, and local safety and health requirements, the most stringent shall apply. Equipment or tools not meeting OSHA requirements will not be allowed on the project sites. Failure to comply with the requirements of this section and related sections may result in suspension of work.

B. Qualifications of Employees:

1. Ensure that employees are physically qualified to perform their assigned duties in a safe manner.
2. Do not allow employees to work whose ability or alertness is impaired because of drugs, fatigue, illness, intoxication, or other conditions that may expose themselves or others to injury.
3. Operators of vehicles, mobile equipment, hoisting equipment, and hazardous plant equipment shall be able to understand signs, signals, and operating instructions, and be capable of operating such equipment. Provide operating instructions for all equipment. Newly hired operators shall be individually tested by an experienced operator or supervisor to determine if they are capable of safely operating equipment.

1.17. ACCIDENT REPORTING:

- A. Reportable Accidents: A reportable accident is defined as death, occupational disease, traumatic injury to employees or the public, property damage by accident in excess of \$100, and fires. Notify Contracting Officer immediately in the event of a reportable accident. Other Accidents: The contractor shall report all other accidents to the Contracting Officer as soon as possible and assist the Contracting Officer and other officials as required in the

PART 2 PRODUCTS

- 2.1 FIRST AID FACILITIES: Provide adequate facilities for the number of employees and the type of construction at the site.
- 2.2 PERSONNEL PROTECTIVE EQUIPMENT: Meet requirements of NIOSH and MSHA, where applicable, as well as ANSI
- 2.3 BARRIERS: Section 01510

PART 3 EXECUTION

- 3.1 EMERGENCY INSTRUCTIONS: Post telephone numbers and reporting instructions for ambulance, physician, hospital, fire department, and police in conspicuous locations at the work site.
- 3.2 EGRESS: Provide and maintain adequate egress at all times in accordance with the Life Safety Code (NFPA 101-94). No corridor, aisle, door, or exit shall be obstructed or used in a manner that interferes with egress
- 3.3 PROTECTIVE EQUIPMENT:
 - A. Inspect personal protective equipment daily and maintain in a serviceable condition. Clean, sanitize, and repair, as appropriate, personal items before issuing them to another individual
 - B. Inspect and maintain other protective equipment and devices before use and on a periodic basis to ensure safe operation.
- 3.4 TRAINING:
 - A. First Aid: Provide adequate training to ensure prompt and efficient first aid.
 - B. Hazardous Material: Train and instruct each employee exposed to hazardous material if any in safe and approved methods of handling and storage. Hazardous materials are defined as explosive, flammable, poisonous, corrosive, oxidizing, irritating, or otherwise harmful substances that could cause death or injury

END OF SECTION

SECTION 012000

PROJECT MEETINGS

PART 1. GENERAL

1.1. PRECONSTRUCTION CONFERENCE

A. Before start of construction, Contracting Officer will arrange a meeting with contractor. The meeting agenda will include the following as a minimum:

1. Correspondence procedures
2. Designation of responsible personnel
3. Labor standards provisions
4. Payroll reports
5. Changes
6. Payments to Contractor
7. Subcontractors
8. National Park Service regulations
9. Accident prevention program (including name of responsible supervisor)
10. Accident reporting
11. Documents required under the contract
12. Park rules and regulations
13. Recycling Program
14. Saturday, Sunday, holiday and night work
15. Tentative construction schedule
16. Submittal of shop drawings, project data, samples, and approved equals
17. Project closeout requirements
18. Relationship of Division 1 to other divisions

END OF SECTION

SECTION 013000

SUBMITTALS

PART 4. GENERAL

4.1. DESCRIPTION: The work of this section consists of submittal requirements before and during construction.

- Ceramic tile color and sizes
- Wall and ceiling paint color; Latex paint in Sherwin-Williams brand or equal and color to be chosen by Park
- Sinks; existing size and stainless steel
- Wall mounted urinal, white porcelain
- Wall mounted toilets; white porcelain
- Vanities; match existing size and shape
- Electric hand dryers; Xlerator brand or equal, Stainless steel color
- Flush valves for urinal and toilets; Zurn brand or equal
- Faucets for sinks; Sloan, Zurn brand or equal in nickel color
- Mirror; size of existing
- Partitions: solid surface material in color chosen by the Park

4.2. SCHEDULES:

- A. Submit four copies of Progress Schedule (normally in bar chart form) showing estimated starting and completion dates for each part of the work. The first progress payment will not be issued until an acceptable progress schedule is submitted.

4.3. SUBMITTAL AND APPROVAL PROCEDURES:

- A. All submittals shall be transmitted using National Park Service form DSC-1(CS). No action will be taken on a submittal item unless accompanied by the transmittal form.
- B. As specified in the individual sections, forward submittals to Contracting Officer at least 20 days before need for approval. Unless a different number is specified, submit one reproducible original and three copies of each shop drawing, four copies of manufacturer's catalog sheets (cut sheets), four specimens of each sample, and four copies of all other submittals requested.
1. Shop Drawings: Not required but recommended. If used ensure they contain:
- a. Date.
 - b. Date of revisions (when applicable).
 - c. Contractor's certification that shop drawing has been checked for compliance with contract documents.
 - d. Details of fabrication, assembly and erection including connections and engagement to contiguous work.
 - e. Materials used.
 - f. All required dimensions.

- g. The term "by others" shall not be used. All work to be performed by others shall be identified by contractor or subcontractor name, discipline, or trade.

2. Samples: Samples shall be large enough to illustrate clearly the functional characteristics and full range of color, texture, or pattern.
 3. Manufacturers' Catalog Sheets: Submit only pertinent pages; mark each copy of standard printed data to identify specific products proposed for use.
 4. Manufacturer's Installation Instructions: When contract documents require compliance with manufacturer's printed instructions, provide one complete set of instructions for Contracting Officer and keep another complete set of instructions at the project site until substantial completion
- C. Contracting Officer reserves the right to require submittals in addition to those called for in individual sections.
- D. Approved Equals:
1. For each item proposed as an "approved equal," submit supporting data, including:
 - a. Drawings and samples as appropriate.
 - b. Comparison of the characteristics of the proposed item with that specified.
 - c. Changes required in other elements of the work because of the substitution.
 - d. Name, address, and telephone number of vendor.
 - e. Manufacturer's literature regarding installation, operation, and maintenance, including schematics for electrical and hydraulic systems, lubrication requirements, and parts lists. Describe availability of maintenance service, and state source of replacement materials.
 2. A request for approval constitutes a representation that contractor:
 - a. Has investigated the proposed item and determined that it is equal or superior in all respects to that specified.
 - b. Will provide the same warranties for the proposed item as for the item specified.
 - c. Has determined that the proposed item is compatible with interfacing items.
 - d. Will coordinate the installation of an approved item and make all changes required in other elements of the work because of the substitution.
 - e. Waives all claims for additional expenses that may be incurred as a result of the substitution
 3. Construction Materials: The contractor is encouraged to submit for approval products made out of recycled or environmentally responsible material. Every effort will be made by the National Park Service to approve these materials.
- E. Coordinate all submittals and review them for legibility, accuracy, completeness, and compliance with contract requirements. Forward submittals that are related to or affect one

another as a package to facilitate coordinated review. Each transmittal shall contain only data specific to that individual submittal.

F. Contracting Officer's Review:

1. Submittals will be returned disapproved without technical review if an incorrect number or format of submittals is provided; the transmittal form is incorrectly filled out; submittals are not coordinated; or submittals do not show evidence of contractor's approval.
2. Any work done or orders for materials or services placed before approval shall be at the contractor's own risk.
3. After reviewing submittals, the Contracting Officer will return one copy of form DSC-1(CS) and one copy of applicable (marked up) submittal sheets to the contractor. Shop drawing review notations will be returned on the reproducible original shop drawing. All other submitted items will be retained. The contractor is responsible for producing additional copies for his/her own use
4. The returned submittal will be marked in one of three ways as defined below:
 - a. APPROVED: Acceptable with no corrections.
 - b. APPROVED WITH NOTATIONS: Minor corrections or clarifications required. All comments are clear and no further review is required. The contractor shall address all review comments when proceeding with the work.
 - c. DISAPPROVED - RESUBMIT: Rejected as not in accordance with the contract or as requiring major corrections or clarifications. The Contracting Officer will identify the reasons for disapproval. The contractor shall revise and resubmit with changes clearly identified.

END OF SECTION

SECTION 01453

ACCESSIBILITY INSPECTIONS AND PROCEDURES

PART 7. GENERAL

- 7.1. DESCRIPTION: The work of this section consists of preparing and executing accessibility inspections and procedures to assure project meets the Architectural Barriers Act Accessibility Standards for Buildings and Facilities (ABAAS).
- 7.2. LAYOUT OF WORK:
 - A. Contractor shall immediately notify the Contracting Officer of apparent errors discovered on the specs. If changes are required, contractor shall cooperate with Contracting Officer in prompt establishment of the field control for altered or adjusted work.
- 7.3. SUBMITTALS:
 - A. As specified in Section 01330
 - B. Submit product submittals and gain approval from COR prior to demolition of existing tile and fixtures.

SECTION 015100

TEMPORARY SERVICES AND CONTROLS

PART 10. GENERAL

- 10.1. DESCRIPTION: The work of this section consists of providing temporary services and controls required for contractor's performance of the work of this contract

PART 11. PRODUCTS

- 11.1. GENERAL: Temporary materials may be new or used, but must be adequate in capacity for the required usage, must not create unsafe conditions, and must not violate requirements of applicable codes and standards.
- 11.2. SANITARY FACILITIES: The park will supply portable toilets for the general public that the contractor is free to use.
- 11.3. FIRE PROTECTION EQUIPMENT: UL Class 2A, 2 1/2 gallon water type stored-pressure extinguisher and UL Class 10, Type I 15-pound B: C carbon dioxide extinguisher.
- 11.4. TAPE: Banner Guard, imprinted with "CAUTION: CONSTRUCTION AREA", manufactured by Reef Industries, Inc., Houston, TX or approved equal.

PART 12. EXECUTION

- 12.1. ELECTRICITY AND LIGHTING: Provide connections to existing facilities. The Government will pay the costs of power used.
- 12.2. TELEPHONE: No telephone service is available on site for contractor's use.
- 12.3. WATER:
- A. Make connections to existing facilities for potable water. Government will pay costs for water used.
 - B. Water for construction is available within the park boundaries.
 - C. Furnish cool, potable water for construction personnel in locations convenient to work stations.
- 12.4. SANITARY FACILITIES:
- A. None needed.

12.5. FIRE PREVENTION AND PROTECTION:

- A. Responsible person: A capable and qualified person shall be placed in charge of fire protection. The responsibilities shall include locating and maintaining fire protective equipment and establishing and maintaining safe torch cutting and welding procedures.
- B. Hazard Control: Take all necessary precautions to prevent fire during construction. Do not store flammable or combustible liquids within 50 feet of a structure. Provide adequate ventilation during use of volatile or noxious substances.
- C. Spark Arresters: Equip all gasoline or diesel powered equipment used during periods of potential fire hazards or in potential forest and grass fire locations with spark arresters approved by the U. S. Forest Service.
- D. Smoking: Smoking within buildings or temporary storage sheds is prohibited. Smoking is prohibited near any combustible materials.
- E. Welding: Cutting by torch or welding shall be performed only when adequate fire protection is provided.

12.6. PROTECTION EQUIPMENT REQUIRED: Travel distance from any work station to the nearest extinguisher shall not exceed 75 feet.

12.7. DISPOSAL:

- A. Unless otherwise specified, all removed material becomes the property of the contractor and shall be disposed of outside the park. All trucks hauling waste and debris shall be covered in such a manner as to guarantee retention of contents.
- B. Immediately remove hazardous rubbish from project site. Place other contraction debris in refuse containers at least daily. Dispose of refuse at least weekly, in a legal manner, at public or private dumping areas outside the park. Do not burn or bury refuse inside the park.

12.8. AIR AND WATER POLLUTION CONTROL:

- A. Take all necessary reasonable measures to reduce air and water pollution by any material or equipment used during construction.

12.9. PROTECTION OF THE PUBLIC:

- A. Ensure safety of the public by providing fencing, barricading, or otherwise blocking off the immediate work area to prevent unauthorized entry.
- B. Erect and maintain barricades, lights, danger signals, and warning signs in accordance with MUTCD.

12.10. HOUSEKEEPING:

- A. Keep project neat, orderly, and in a safe condition at all times.
- B. Provide enough containers for collecting construction debris and construction materials to be recycled.
- C. Keep volatile wastes in covered containers.

END OF SECTION

SECTION 01600

MATERIALS AND EQUIPMENT

PART 13. GENERAL

- 13.1. DESCRIPTION: The work of this section consists of the general procedures for handling, storing, and protecting material and equipment.
- 13.2. TRANSPORTATION AND HANDLING: Arrange deliveries of materials in accordance with construction schedules; coordinate to avoid conflict with work and conditions at the site. Deliver materials in undamaged condition, in manufacturer's original containers or packaging, with identifying labels intact and legible. Contractor is encouraged to obtain materials in biodegradable or recyclable/reusable packaging, which uses the minimum amount of packaging possible.
- 13.3. STORAGE AND PROTECTION:
- A. Store materials in accordance with manufacturer's instructions, with seals and labels accessible for inspection.
 - B. Storage:
 - 1. Store products subject to damage by the elements in weather tight enclosures.
 - 2. Store fabricated products above the ground, on blocking or skids; prevent soiling or staining. Cover products subject to damage or deterioration with impervious sheet coverings; provide adequate ventilation to avoid condensation.
 - 3. Store loose granular materials in a well-drained area on solid surfaces to prevent mixing with foreign matter.
 - C. Protection after Installation: Provide adequate coverings as necessary to protect installed materials from damage resulting from natural elements, traffic, and subsequent construction. Remove when no longer needed.

END OF SECTION

SECTION 01700

PROJECT CLOSEOUT

PART 16. GENERAL

- 16.1. DESCRIPTION: The work of this section consists of final cleanup and final inspection procedures.

PART 17. PRODUCTS

- 17.1. CLEANING MATERIALS: As recommended by the manufacturer of surface to be cleaned.

PART 18. EXECUTION

- 18.1. CLEANING: Before scheduling the final inspection, remove all tools, equipment, surplus materials, and rubbish. Restore or refinish surfaces that are damaged due to work of this contract to original condition. Remove grease, dirt, stains, foreign materials, and labels from finished surfaces. Pick up all construction debris from the site. At time of final inspection, project shall be thoroughly clean and ready for use.

18.2. PROJECT RECORD DRAWINGS

- A. Keep record drawings current. Make record drawings available to the Contracting Officer for inspection at the time of monthly progress payment requests. If project record drawings are not current, the Contracting Officer may retain an appropriate amount of the progress payment.
- B. On completion of the total project, submit complete record drawings. Include all shop drawings, sketches, and additional drawings that are to be included in the final set, with clear instructions showing the location of these drawings.

18.3. CLOSEOUT SUBMITTALS:

- A. Project Record Drawings: As specified above.
- B. Spare Parts and Materials: As specified in individual sections.
- C. Operation and Maintenance Data: As specified in individual sections.
- D. Mechanical and Electrical Systems: Verify the following in writing:

1. All systems are complete, properly started, and are operational.
 2. All controls are complete and operational and sequences have been checked and are functioning properly.
- 18.4. SUBSTANTIAL COMPLETION AND FINAL INSPECTION: Submit written certification that project, or designated portion of project, is substantially complete, and request in writing a final inspection. Contracting Officer Representative will make an inspection within 10 days of receipt of request.
- A. When work is determined to be substantially complete, Contracting Officer will prepare a list of deficiencies to be corrected before final acceptance and issue a Letter of Substantial Completion.
 - B. If work is not determined to be substantially complete, Contracting Officer will notify contractor in writing. After completing work, contractor shall resubmit certification and request a new final inspection.
- 18.5. ACCEPTANCE OF THE WORK: After all deficiencies have been corrected, a Letter of Acceptance will be issued.

END OF SECTION

SECTION 017850

OPERATION AND MAINTENANCE DATA

PART 19. GENERAL

- 19.1. DESCRIPTION: The work of this section consists of furnishing operation and maintenance data manuals.
- 19.2. RELATED WORK: Project Record Drawings- Section 01700
- 19.3. SCHEDULING:
 - A. At start of project, begin accumulating operation and maintenance data and initiate an index. Install and index all data in binders within 30 days after delivery of items. As custom written data and test results are produced, add them to the operation and maintenance data file.
 - B. Keep operation and maintenance data current. Make operation and maintenance binders available to the Contracting Officer for inspection at the time of monthly progress payment requests. If operation and maintenance binders are not current, the Contracting Officer may retain an appropriate amount of the progress payment.
 - C. Before scheduling a final inspection, furnish two complete sets of operation and maintenance data to Contracting Officer for review. Should Contracting Officer find manual to be substantially incomplete, the final inspection will be delayed.
 - D. Within 30 days following receipt of review comments, deliver two completed sets of Operation and Maintenance data.

PART 20. PRODUCTS

- 20.1. BINDERS: White, commercial quality, hard back, three-ring, 2-inch maximum ring size, lever-locking type slant ring, with clear window pockets on front and side.
- 20.2. INDEX SYSTEM: Index sheet with mylar reinforced edges at binder holes and tabbed divider sheets with mylar reinforced edges and pre-printed numbered tabs aligned with numbers and title lines on index sheet.

21.2. CONTENTS

- A. **Manufacturers' Published Data:** Provide all available data, including installation and operating instructions, parts lists, electrical and mechanical schematics, control circuit documentation, performance data, safety instructions, cleaning and care instructions, and illustrations and instructions for maintenance, including lubrication, disassembly and repair, cleaning, and service. Indicate catalog numbers, sizes, colors, options, and other information pertaining to the products furnished which would be required when ordering replacements. For equipment assemblies, provide data for each separate item of equipment furnished as part of the assembly
- B. **Custom Written Data:** For data not in manufacturer's standard literature, provide text, drawings, and schematics specifically applicable to installed systems. Include step-by-step descriptions of operating procedures; identification of individual components and their functions; descriptions of how system components relate to one another and operate together to accomplish a common process or function; and sequence of operation for system control circuits. For seasonally operated systems, provide start-up and shutdown instructions.
- C. **Equipment Data Sheets:** For each item of equipment included in the operation and maintenance data, provide an Equipment Data Sheet using the form at the end of this section. For equipment consisting of both a driven machine and a driver (for example, a pump and a motor), the equipment data shall cover both the driven machine and the driver. For similar type equipment (for example, multiple exhaust fans of the same model and type), provide a single equipment data sheet with an attached schedule listing the individual equipment items.
- D. **Vendor Furnished As-Built Drawings:** Provide for each electrical and mechanical control system.
 - 1. For each control system, provide control circuit schematic drawings. Identify each wire and terminal block number. Show terminal numbers on all control devices. Show control wires and devices remote from the control panel.
 - 2. For each control panel, provide a general arrangement drawing showing location of each control component and terminal block on the panel front and interior. Include a materials list of all panel-mounted control components as well as field-installed control components remote from the panel, identifying components, manufacturer, model number, and initial set points or sensing ranges of devices where applicable
- E. **Schedules:** Provide one copy of material and equipment schedules as applicable
- F. **Warranties:** Place a copy of each manufacturer, supplier, and installer warranty extending for a period greater than one year in a single separately identified tabbed section of the manual
- G. **Test Results:** Include in the operation and maintenance data copies of test results for electrical equipment and systems as listed in the individual specification sections

- H. Subcontractor and Supplier List: List all subcontractors and major suppliers who worked on the project. Include each subcontractor's or supplier's address and telephone number and identify work performed.
- I. List of sections requiring operation and maintenance data:
 - 1. 16010 – General Electrical Provisions
 - 2. 16100 – Basic Materials and Methods
- J. Subcontractor and Supplier List\ : List all subcontractors and major suppliers who worked on the project. Include each subcontractor's or supplier's address and telephone number and identify work performed

END OF SECTION

